# HAVN Rest Cabins

Health, sleep and longevity concept. Technical summary for Soundbox Inc., Guangzhou. HAVN, Berlin, 21 October 2026. Basis for discussion, not a specification release.

## 1. Cabin types

Every HAVN station has 11 freestanding room-in-room cabins and a shared sanitary core (showers, WC, lockers). All three types use the same panel system and the same soundproof door: 2.2 m high, flat-pack panels max. 80 cm wide, fresh-air ventilation, 230 V EU power, CE, reaction-to-fire class per EN 13501-1.

| Type | Per station | Floor (m) | Bed / interior | Use |
|---|---|---|---|---|
| Single Cabin | 5 | 2.0 x 2.4 | Bed 200 x 100 cm, small desk and chair, hooks, luggage space under the bed | 2 h, 4 h, night (8 h), day pass (12 h) |
| Double Cabin | 4 | 2.4 x 2.4 | Bed 200 x 160 cm, hooks and luggage space for two | 2 h, 4 h, night, day pass |
| Health Pod | 2 | 2.0 x 2.4 | Therapy bed, wall-mounted red/NIR light panel (630-850 nm), audio, scent; no sleeping bed | Per hour; guided 20-minute protocols |

## 2. Design principle: four measurable conditions plus one light sequence

The health value of a rest cabin lies in conditions that can be measured and documented per stay: darkness, quiet, fresh air and the right temperature, plus a disciplined light sequence (warm low light in the evening, true dark at night, a 30-minute dawn ramp before waking). These levers carry the strongest evidence, they are inexpensive, and the best sleep operators (Zedwell London, Equinox New York) sell exactly this set. Everything else is comfort or an opt-in extra.

| Parameter | HAVN target | Basis |
|---|---|---|
| Noise at the pillow | LAeq,night at most 30 dB(A). Envelope to DIN 4109 enhanced level: wall R'w at least 52 dB, door Rw at least 37 dB; silencers on both ventilation penetrations; measured dB(A) at the pillow to be published | WHO bedroom guideline; DIN 4109 Beiblatt 2. Office pods rated under ISO 23351-1 reach about 30 dB; a station site needs roughly 10 dB more |
| Fresh air / CO2 | At least 36 m3/h per person (72 m3/h in the Double). ERV idles at about 20 % and boosts at 750 ppm CO2; alarm at 1,000 ppm; full-flow purge during every reset. HEPA + activated carbon on the supply. No ioniser, plasma or ozone device | EN 16798-1 Cat. I; DTU chamber studies (sleep efficiency and awakenings worsen above 1,000 ppm); WELL v2 |
| Temperature | Guest setpoint 18-24 C, default 20 C. Sensible cooling capacity about 300-600 W per cabin (water loop or small split) with heat rejection into the host building; heavier-duvet option; bed-level cooling pad optional | Baniassadi et al. 2023 (about 11,000 person-nights); sleep-medicine guidance |
| Light during sleep | Below 1 lux at the pillow; all status LEDs off or deep red; hard-off relay for the virtual window and the control tablet | Brown et al. 2022 consensus; PNAS 2022 (100 lux at night raises heart rate and next-day insulin resistance) |
| Particulates | PM2.5 at most 10 ug/m3 on the supply air | WELL v2 A05 |
| Humidity | 40-60 % RH (design target) | Conventional practice |

## 3. Lighting: a four-channel circadian system

Tunable white (2200-6500 K, CRI 90 or higher) plus a separate amber/red channel on a 24 V system, flicker-free drivers (PstLM at most 1.0, SVM at most 0.4 as in EU Ecodesign 2019/2020), control by DALI-2 DT8 or Casambi (e.g. Sunricher or LEDGEAR components), scenes executed locally on the cabin controller. A tunable-white strip alone cannot deliver usable night light below 1 lux melanopic, which is why the amber/red channel is needed. HAVN will ask for the measured spectral power distribution and the computed melanopic EDI per channel, not a CCT figure alone. Every light source needs an EN 62471 photobiological report.

| Scene | Melanopic EDI at the eye | CCT | Photopic lux at the eye |
|---|---|---|---|
| Daytime arrival, reading, nap exit | at least 250 lx | 4000-6500 K | about 265-350 lx |
| Evening arrival (last 3 h before sleep) | at most 10 lx | 2200-2700 K | about 20 lx ambient, warm localised reading light |
| Wind-down (last 20-30 min) | 1-5 lx | amber, below 2200 K | a few lux |
| Sleep | at most 1 lx | - | below 1 lx; window and tablet hard-off |
| Night navigation | at most 10 lx | deep red / amber at floor level | 5-20 lx of long-wavelength light as floor cue lines |

| Scene | Melanopic EDI at the eye | CCT | Photopic lux at the eye |
|---|---|---|---|
| Dawn wake (night stays) | ramp to 100-250 lx | 2000 K to 4000-5000 K | 0 to 100-300 lx over 30 minutes, starting before the alarm |

## 4. The virtual window

HAVN cabins have no real window. Randomised studies show that an artificial window or skylight lowers tension, anxiety and claustrophobia and improves mood in windowless rooms; no study yet covers sleeping rooms, so HAVN will measure this in the pilot. Depth is the decisive filter for a 100 mm panel:

| Technology / product | Depth | Power | Fits a 100 mm wall? | Price indication |
|---|---|---|---|---|
| Innerscene Circadian Sky (tunable sky, no sun), 300 x 600 to 600 x 1200 mm | 84 mm | 35-75 W | Yes, flush | On quotation |
| Edge-lit "virtual window" / skylight light box (e.g. Sky Factory type, Chinese OEM), 450 x 1200 or 600 x 1200 mm | about 45-85 mm | tens of W | Yes, surface or flush; also ceiling | OEM about USD 100-600 per module |
| Thin LCD window (e.g. Atmoph 27") | 30-60 mm | n/a | Yes | USD 380-450 |
| Rayleigh sun-and-sky units (Yeelight Pro P21, Mitsubishi Misola, Innerscene A7, CoeLux HT25) | 120-270 mm | 80-200 W | No: roof cut-out with plenum above the cabin only; host ceiling above 2.5 m | EUR 373 (Yeelight) to USD 6,000-17,400 |

- A portrait 450 x 1200 mm panel fits inside one 80 cm wall module; a landscape 1200 x 600 spans two modules and puts a panel joint behind the luminaire.
- Recessing an 84 mm luminaire into a 100 mm acoustic sandwich removes mass and absorber over that area and creates an acoustic weak spot. HAVN therefore prefers the ceiling panel (where the "outside" is the host room) or a surface-mounted 46 mm light box on the wall.
- The window runs on the same lighting control: daytime light dose, dawn simulator in the morning, hard-off during the sleep window. Its heat (up to 200 W for sun-and-sky units) must be carried by the ventilation stream.
- For EU sale an OEM panel needs CE (Low Voltage and EMC), RoHS, an EN 62471 report and flicker data (PstLM, SVM).

## 5. The Health Pod

Same shell, therapy bed instead of a sleeping bed, wall-mounted red/near-infrared panel, audio and scent. Positioning: wellness and recovery support, explicitly not a medical device. HAVN will not make skin, sleep-disorder, disease or "longevity" claims for the panel; the best-supported effects of red/NIR light are muscle recovery, mood and alertness, so sessions are scheduled in the daytime, not before sleep.

- Protocols: 20-minute power nap (settling with a breathing cue, near-dark, lying time capped, wake by rising audio and brightening light, 2-3 minutes of bright "re-boot" light before exit); 10-minute resonance-breathing or guided rest audio; red-light session of about 10 minutes followed by 10 minutes of rest; 90-minute full sleep cycle with a soft movement-based wake window.
- Panel engineering: half-body panel (about 91 x 30 cm, 300-500 W) fixed on the long wall at 20-30 cm from the bed; nine-point irradiance map at mounting distance from a calibrated spectroradiometer; EN 62471 report including the separate retinal-thermal limit for 780-1400 nm; session timer and per-booking exposure lock; thermal cut-out; disposable eye protection; no flashing or strobe modes.
- Heat: a 450 W panel running 15 minutes releases about 400 kJ into a 10.6 m3 cabin. The ventilation must boost to at least 100 m3/h during and after a session, and fan noise of the panel must be specified.
- Optional later modules: vibroacoustic transducers in the therapy bed, far-infrared warm mat with surface temperature capped around 40-45 C, massage bed. Pulsed electromagnetic field mats, compression boots and cold exposure are not planned for unattended use.

## 6. Sensing and controls

- One contactless 60 GHz radar module per cabin (breathing, heart rate, movement, bed occupancy; about EUR 50 per module at single quantity) doubles as presence sensor for the automatic reset. No cameras.
- Environment sensors: CO2/temperature/humidity (e.g. Sensirion SCD41), particulates (SPS30), sound level, light level. Per-stay environment record (CO2, temperature, dB) stored by the cabin controller.
- Controller: local scene execution (light, ventilation, temperature, audio, scent, window), no cloud dependency, API to the HAVN booking system; platform to be agreed (Tuya Zigbee hotel RCU, ESP32-based, or Matter).

– Guest-facing functions: optional sleep report after the stay (opt-in per stay), soft wake window in light sleep, in-app health screening before Health Pod modules. Data processing follows EU GDPR: tracking off by default, explicit consent, pseudonymised stay IDs, EU hosting.

## 7. Materials, hygiene and compliance checklist

These are the documents HAVN needs from the cabin manufacturer for the Berlin pilot and for series production in the EU.

| Area | Requirement | Standard / rule |
|---|---|---|
| Formaldehyde | Board materials ENF or E0 (GB/T 39600-2021). Chamber test report on the finished cabin article at or below 0.062 mg/m3 from an ISO 17025 laboratory. Chinese E1 boards (0.124 mg/m3) no longer meet the EU limit in force since 6 August 2026 | REACH Annex XVII entry 77, Appendix 14 |
| VOC emissions | 28-day emission test on the assembled wall, ceiling and floor build-ups (not on raw boards); REACH SVHC declaration for foams, adhesives and textiles | German AgBB scheme, 2024 edition |
| Reaction to fire | Classification reports from an EU-accredited laboratory (GB 8624 reports are not accepted in Germany): B-s1,d0 for every exposed lining, Cfl-s1 for the floor; EN 1021 ignitability for upholstery | EN 13501-1; EN 1021 |
| Fire safety provisions | Smoke-detector mount wired to the building alarm; provision for a sprinkler drop head or pass-through in the roof panel; door opens from inside without key or power at all times, electronic lock releases on power loss; main isolator reachable from outside | German building law (MBO / BauO Berlin), VdS CEA 4001:2024, host-building fire concept |
| Acoustics | Laboratory reports for the complete cabin and for the door; sound power of the ventilation unit at low and full speed | EN ISO 10140; ISO 23351-1 |
| Electrical / CE | CE under the Low Voltage Directive 2014/35/EU, EMC Directive 2014/30/EU, RoHS; Radio Equipment Directive for Wi-Fi, Bluetooth or NFC modules. CB-scheme reports (IEC 60335 / 62368-1 / 60598) from TUV, SGS or Intertek; EMC test of the complete cabin with all devices running; bill of electrical components with their own declarations; Schuko wiring | EN 62368-1, EN 55032/55035, EN IEC 61000-3-2/-3-3, EN 60598 |
| Light sources | Photobiological safety report for every light source, including the red/NIR panel (with IR retinal-thermal class) and the virtual window; flicker data PstLM and SVM | EN 62471; Ecodesign 2019/2020 |
| Air treatment | No ionisers, "plasma" or ozone devices. UV-C surface disinfection (254 nm, ozone-free) only while the cabin is locked and with a door interlock | UBA indoor-air commission; Directive 2006/25/EC; IEC 60335-2-65 |
| Surfaces and textiles | PVC-free and phthalate-free surfaces; zipped waterproof washable mattress and pillow encasements; wipeable non-fabric surfaces for all shared contact items in the Health Pod; antimicrobial additives only with approved actives and labelling (or none) | EU Biocidal Products Regulation |
| Reset | Radar confirms the cabin is empty; full-flow air purge for the whole 20-minute reset; linen cassette exchangeable in under five minutes | HAVN operating concept |

## 8. Priorities per cabin type

| Tier | Single and Double Cabin | Health Pod |
|---|---|---|
| Must-have for the Berlin pilot (Nov 2026, 11 cabins) | Envelope and door to the acoustic targets with duct silencers; true blackout with hard-off relays; CO2-controlled ventilation with HEPA + carbon; temperature setpoint and cooling path; four-channel lighting with dawn ramp and floor-level red cue lines; mattress encasement and linen cassette; radar presence sensor; sensor and controller cabling; prepared ceiling or wall position for the virtual window; the complete documentation of section 7. Double: ventilation sized for two, two reading lights | Single Cabin baseline plus: therapy bed; panel mounting frame at fixed distance; irradiance map and EN 62471 report; session timer and interlock; ventilation boost of at least 100 m3/h; audio and light for the 20-minute protocols; wipeable surfaces; manufacturer's declared non-medical intended purpose |
| Should-have for series production | Virtual window (Circadian Sky 450 x 1200 portrait or OEM edge-lit light box) scheduled as daylight dose and dawn simulator; pink-noise masking speaker; lavender cartridge diffuser; bed-level cooling pad (dual-zone in the Double); weighted blanket as booked extra; opt-in sleep report; interlocked UV-C reset module; cove uplight and light matte ceiling | Breathing and guided-rest audio library; bright nap-exit light; vibroacoustic transducers in the bed; far-infrared warm mat with temperature cap; OEM 300 W-class panel once the compliance package is proven |
| Optional / experimental | Rayleigh sun-and-sky skylight in a roof cut-out where host ceilings exceed 2.5 m; pilot measurement window vs. no window | Massage bed; PEMF mat only with screening gate; no cold exposure, oxygen or grounding products |

## 9. Questions for Soundbox

1. Materials: which formaldehyde grade do your boards carry (E1 / E0 / ENF under GB/T 39600-2021), what are core, laminate, foam, felt and floor made of, and can you provide the REACH Appendix 14 chamber report and an AgBB VOC test on the assembled build-ups?
2. Fire: which fire reports do you hold today (GB 8624 or EN 13501-1 from an EU laboratory), and can you reach B-s1,d0 linings and Cfl-s1 floor?
3. Acoustics: what Rw do your wall and door achieve (EN ISO 10140-2) and which ISO 23351-1 class? What would it take to reach 52 / 37 dB with both ventilation penetrations silenced?
4. Ventilation: which Panasonic ERV model is specified (heat-recovery efficiency, CO2-demand input, HEPA + carbon filter option, power and sound power at 36 and 180 m3/h)?
5. Integration: can an 84 mm luminaire or a 46 mm light box sit within one 80 cm panel module without losing acoustic performance, and can the roof panel take a 200-270 mm cut-out? Do you have Guangdong partners for artificial-sky or virtual-window panels?
6. Lighting and controls: four-channel lighting on DALI-2 DT8 or Casambi, with measured spectra, melanopic values and flicker data per channel? Which controller platform and API do you propose for the booking system?
7. Red-light panel: supplier, chip count, drive current, nine-point irradiance map at 20 cm, EN 62471 report including the IR retinal-thermal class, fan noise, surface temperature, timer and interlock design, declared intended purpose.
8. CE: will Soundbox supply CB-scheme reports, the whole-cabin EMC test and component declarations so that HAVN can hold the EU technical file as manufacturer of record?
9. Cooling: which cooling do you integrate (split, water loop, Peltier) for 300-600 W sensible load, and what is the heat-rejection path into the host building?
10. Fire provisions: ceiling provision for sprinkler drop heads or pass-through, smoke-detector mount, fail-safe door release on power loss, external main isolator?
11. Reset: mattress encasement and linen-cassette fixing, interlocked UV-C module, radar occupancy sensor, controller platform and API?
12. Prototype: price and lead time for one Single Cabin and one Health Pod as described, and indicative price breaks at 11, 100 and 500 cabins per type.

## 10. Selected references


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